

Cybersecurity Incident Analysis

ReportProcess, Memory, and Network Correlation Investigation

Introduction

This report presents an analysis of a simulated Windows process log environment. The objective of the investigation is to identify potential malicious activity through process behavior analysis, memory inspection indicators, and network communication patterns.

The analysis focuses on detecting abnormal process relationships, signs of memory injection, and possible command-and-control (C2) communication.

2. Process Analysis

The system shows a normal Windows startup chain:

smss.exe - wininit.exe - services.exe - svchost.exe - winlogon.exe - explorer.exe

However, an anomaly was detected:

notepad.exe (PID 2550) spawned

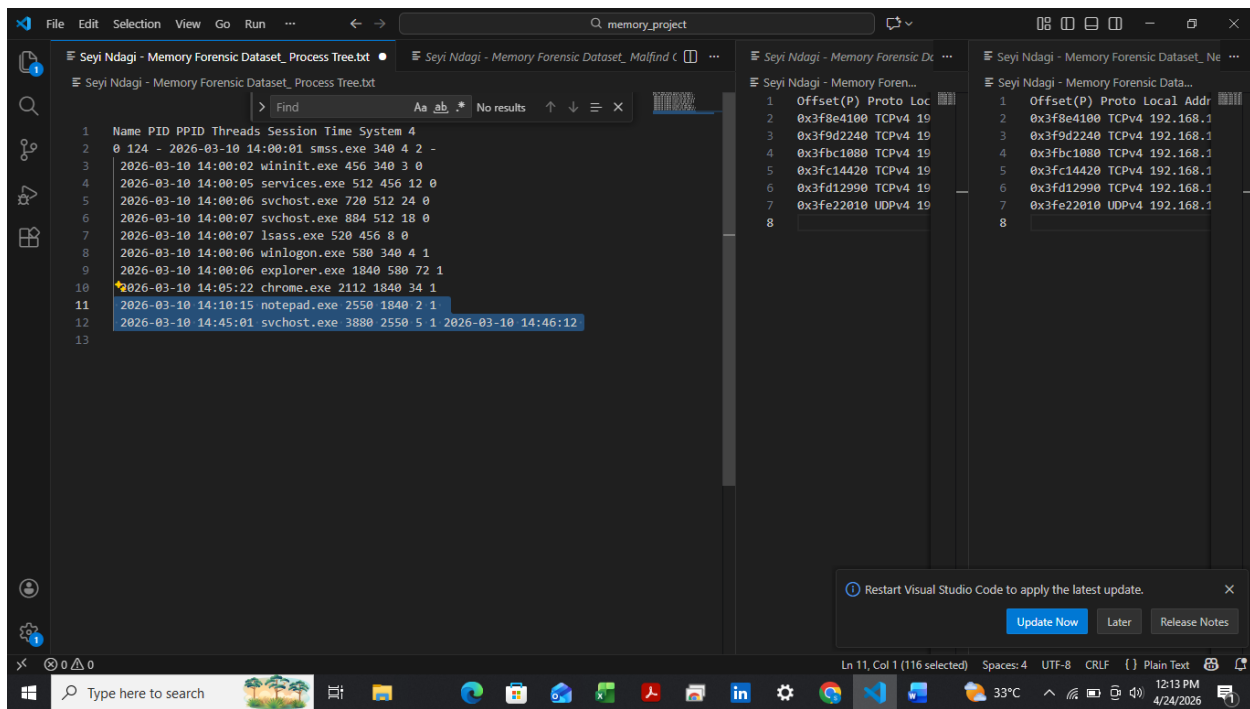
svchost.exe (PID 3880)

This violates normal Windows process hierarchy, where svchost.exe should be spawned by services.exe, not a user application.

3. Suspicious Process

svchost.exe (PID 3880)

This indicate Abnormal parent process (notepad.exe), Unexpected execution behavior and Correlated network activity

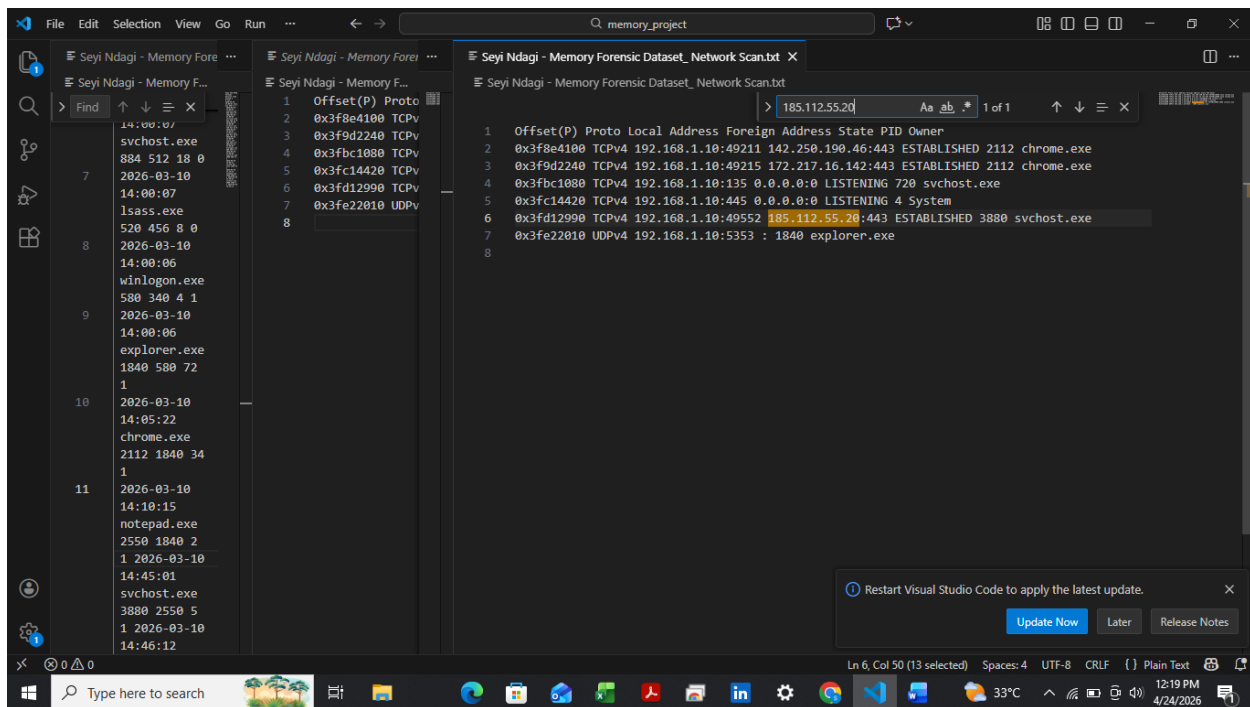


4. Network Activity

Destination IP: **185.112.55.20**

Port: **443 (HTTPS)**

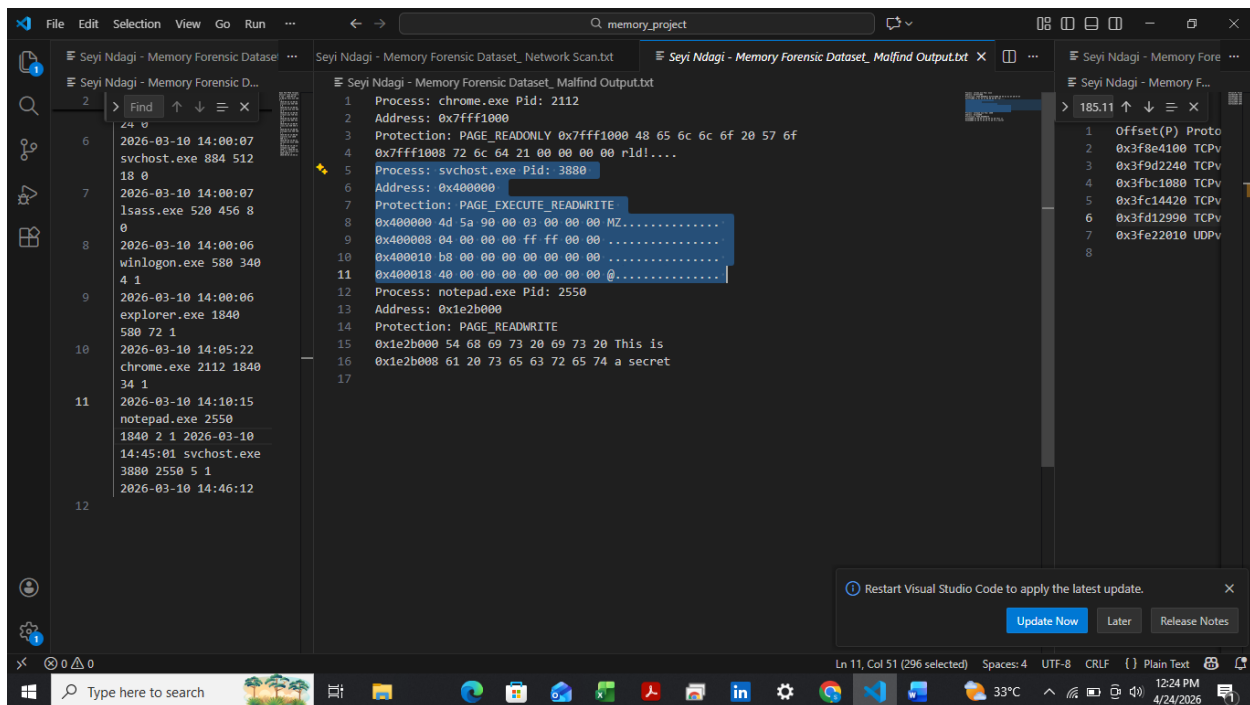
The process communicates externally over HTTPS, which may indicate concealed command and control (C2) traffic.



5. Memory Analysis

RWX memory region detected and Presence of MZ header

This indicates executable code injected into process memory, consistent with fileless malware behavior.



6. Conclusion

The combined evidence suggests Abnormal process spawning behavior, Memory injection into svchost.exe and External network communication.

This activity is consistent with a **process injection / fileless malware scenario** using svchost.exe for stealth execution and communication.

Author: Mohammed Ndagi, Date: April 2026.